Lecture 2

Nilpotency and Engel's theorem

Objectives. *Iterated brackets define nilpotency of an abstract Lie algebra. Strictly upper triangular matrices provide the model for nilpotency. Engel's theorem recognises this behaviour from the adjoint action, through nilpotent operators. We prove the linear form first. We then apply it to* ad *and follow the proof in the strictly upper triangular* 4×4 *matrices.*

By the end of this lecture you will be able to:

- Compute the lower central series of a matrix Lie algebra.
- Reconstruct the common-null-vector induction in Engel's theorem.
- Deduce the abstract form of Engel's theorem through ad.
- Distinguish a nilpotent Lie algebra from an algebra of nilpotent matrices.

Vector spaces and Lie algebras are finite-dimensional over $\mathbb{C}$. A linear map $z : V \to V$ is *nilpotent* if $z^m = 0$ for some $m \geq 1$.

2.1 The lower central series

As in Lecture 1, $[U, W]$ denotes the span of all brackets $[u, w]$ with $u \in U$ and $w \in W$.

Definition 2.1 (Lower central series and nilpotency)**.** The *lower central series* of $\mathfrak{g}$ consists of the subspaces

$$C^0\mathfrak{g} = \mathfrak{g}, \qquad C^{k+1}\mathfrak{g} = [\mathfrak{g}, C^k\mathfrak{g}] \qquad (k \geq 0).$$

The Lie algebra $\mathfrak{g}$ is *nilpotent* if $C^m\mathfrak{g} = 0$ for some $m \geq 0$.

Thus $C^1\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$. By bilinearity of the bracket and induction on k, the subspace $C^k\mathfrak{g}$ is spanned by the nested brackets

$$[x_1, [x_2, \ldots, [x_k, y] \ldots]] = (\operatorname{ad} x_1)(\operatorname{ad} x_2) \cdots (\operatorname{ad} x_k)(y), \qquad x_1, \ldots, x_k, y \in \mathfrak{g}.$$

Lemma 2.2. *Let $\mathfrak{g}$ be a Lie algebra and $k \geq 0$.*

(i) $C^{k+1}\mathfrak{g} \subseteq C^k\mathfrak{g}$, and $C^k\mathfrak{g}$ is an ideal of $\mathfrak{g}$.

(ii) If $\mathfrak{s} \subseteq \mathfrak{g}$ is a subalgebra, then $C^k\mathfrak{s} \subseteq C^k\mathfrak{g}$.

(iii) If $\varphi : \mathfrak{g} \to \mathfrak{k}$ is a surjective homomorphism, then $\varphi(C^k\mathfrak{g}) = C^k\mathfrak{k}$.

Hence subalgebras and quotients of nilpotent Lie algebras are nilpotent.

Proof. (i) First, $C^1\mathfrak{g} \subseteq \mathfrak{g} = C^0\mathfrak{g}$. If $C^k\mathfrak{g} \subseteq C^{k-1}\mathfrak{g}$, then

$$C^{k+1}\mathfrak{g} = [\mathfrak{g}, C^k\mathfrak{g}] \subseteq [\mathfrak{g}, C^{k-1}\mathfrak{g}] = C^k\mathfrak{g}.$$

Thus $[\mathfrak{g}, C^k\mathfrak{g}] = C^{k+1}\mathfrak{g} \subseteq C^k\mathfrak{g}$. This is the ideal condition.

(ii) The case $k = 0$ is $\mathfrak{s} \subseteq \mathfrak{g}$. Inductively,

$$C^{k+1}\mathfrak{s} = [\mathfrak{s}, C^k\mathfrak{s}] \subseteq [\mathfrak{g}, C^k\mathfrak{g}] = C^{k+1}\mathfrak{g}.$$

(iii) The case $k = 0$ is surjectivity. Assume that $\varphi(C^k\mathfrak{g}) = C^k\mathfrak{k}$. Since φ is linear and preserves brackets,

$$\varphi(C^{k+1}\mathfrak{g}) = \operatorname{span}\{[\varphi(x), \varphi(c)] : x \in \mathfrak{g},\ c \in C^k\mathfrak{g}\} = [\mathfrak{k}, C^k\mathfrak{k}].$$

The last step uses $\varphi(\mathfrak{g}) = \mathfrak{k}$ and this inductive hypothesis.

For the final sentence, apply (ii) to a subalgebra and (iii) to a quotient map $\mathfrak{g} \to \mathfrak{g}/\mathfrak{a}$. ∎

Lemma 2.3 (Centre and central quotients). *Let $\mathfrak{g}$ be a Lie algebra.*

(i) If $\mathfrak{g} \neq 0$ is nilpotent, then $Z(\mathfrak{g}) \neq 0$.

(ii) If $\mathfrak{g}/Z(\mathfrak{g})$ is nilpotent, then $\mathfrak{g}$ is nilpotent.

Proof. (i) Let m be the least integer with $C^m\mathfrak{g} = 0$. Since $\mathfrak{g} \neq 0$, we have $m \geq 1$. By minimality, $C^{m-1}\mathfrak{g} \neq 0$. Moreover,

$$[\mathfrak{g}, C^{m-1}\mathfrak{g}] = C^m\mathfrak{g} = 0.$$

Hence $C^{m-1}\mathfrak{g}$ lies in $Z(\mathfrak{g})$.

(ii) Let $q\colon \mathfrak{g} \to \mathfrak{g}/Z(\mathfrak{g})$ be the quotient map, and suppose $C^m(\mathfrak{g}/Z(\mathfrak{g})) = 0$. Lemma 2.2(iii) gives $q(C^m\mathfrak{g}) = 0$, so $C^m\mathfrak{g} \subseteq Z(\mathfrak{g})$. Therefore

$$C^{m+1}\mathfrak{g} = [\mathfrak{g}, C^m\mathfrak{g}] \subseteq [\mathfrak{g}, Z(\mathfrak{g})] = 0.$$ ∎

Example 2.4 (Three earlier algebras). An abelian algebra has $C^1\mathfrak{g} = 0$. For the Heisenberg algebra of Lecture 1,

$$C^1\mathfrak{h}_3 = \mathbb{C}z, \qquad C^2\mathfrak{h}_3 = [\mathfrak{h}_3, \mathbb{C}z] = 0,$$

so $\mathfrak{h}_3$ is nilpotent. For the upper triangular algebra $\mathfrak{b} \subseteq \mathfrak{gl}_2(\mathbb{C})$, we know $C^1\mathfrak{b} = \mathbb{C}e$. Since $\mathbb{C}e$ is an ideal and $[h, e] = 2e$, induction gives

$$C^{k+1}\mathfrak{b} = [\mathfrak{b}, \mathbb{C}e] = \mathbb{C}e \quad (k \geq 1).$$

Hence $\mathfrak{b}$ is not nilpotent. Finally, $[\mathfrak{sl}_2, \mathfrak{sl}_2] = \mathfrak{sl}_2$ gives $C^k\mathfrak{sl}_2 = \mathfrak{sl}_2$ for every k.

The matrix unit E_{ij} has entry 1 in position (i, j) and zero elsewhere. The product rule for matrix units is

$$E_{ij}E_{kl} = \delta_{jk}E_{il}.$$

Definition 2.5 (Strictly upper triangular matrices). For $n \geq 1$, let $\mathfrak{n}_n \subseteq \mathfrak{gl}_n(\mathbb{C})$ be the span of the E_{ij} with $i < j$. For $r \geq 1$, let

$$N_r = \operatorname{span}\{E_{ij} : j - i \geq r\}.$$

Thus $N_1 = \mathfrak{n}_n$. Since $j - i \leq n - 1$, we have $N_r = 0$ for $r \geq n$. The integer $j - i$ is the height of the entry above the diagonal.

Lemma 2.6 (Height lemma). *For $r, s \geq 1$,*

$$N_rN_s \subseteq N_{r+s}, \qquad [N_r, N_s] \subseteq N_{r+s}.$$

Consequently $\mathfrak{n}_n$ is a subalgebra, $C^k\mathfrak{n}_n \subseteq N_{k+1}$, and $C^{n-1}\mathfrak{n}_n = 0$.

Proof. Let $j - i \geq r$ and $l - k \geq s$. The product $E_{ij}E_{kl}$ vanishes unless $j = k$. In that case it equals E_{il}, and

$$l - i = (l - k) + (j - i) \geq r + s.$$

This gives the product inclusion in both orders, hence the bracket inclusion. The case $r = s = 1$ says that $\mathfrak{n}_n$ is a subalgebra. Starting from $C^0\mathfrak{n}_n = N_1$, induction gives

$$C^{k+1}\mathfrak{n}_n = [N_1, C^k\mathfrak{n}_n] \subseteq [N_1, N_{k+1}] \subseteq N_{k+2}.$$

Finally, $C^{n-1}\mathfrak{n}_n \subseteq N_n = 0$. ∎

For $n = 3$, put $p = E_{12}$, $q = E_{23}$ and $z = E_{13}$. These are the matrices of Lecture 1, so $\mathfrak{n}_3 = \mathfrak{h}_3$.

2.2 Engel's theorem for nilpotent matrices

Lemma 2.7. *Let $m \geq 1$. If $x \in \mathfrak{gl}(V)$ satisfies $x^m = 0$, then $(\operatorname{ad} x)^{2m-1} = 0$ on $\mathfrak{gl}(V)$.*

Proof. Define endomorphisms λ and r of $\mathfrak{gl}(V)$ by

$$\lambda(y) = xy \quad \text{and} \quad r(y) = yx.$$

Associativity gives $\lambda r(y) = xyx = r\lambda(y)$, so these maps commute. Also $\operatorname{ad} x = \lambda - r$. The binomial theorem for commuting operators gives

$$(\operatorname{ad} x)^{2m-1}(y) = \sum_{i=0}^{2m-1} \binom{2m-1}{i} (-1)^{2m-1-i}\, x^i\, y\, x^{2m-1-i}.$$

In each term, $i \geq m$ or $2m - 1 - i \geq m$. Hence every term is zero. ∎

A nilpotent operator z on a nonzero space U kills a nonzero vector. Indeed, take $u \neq 0$ and let j be the largest integer with $z^j u \neq 0$. Then $z^j u$ is a nonzero vector in $\ker z$.

Theorem 2.8 (Engel's theorem, linear form)**.** *Let $V \neq 0$, and let $L \subseteq \mathfrak{gl}(V)$ be a Lie subalgebra whose elements are all nilpotent. Then there is a vector $v \neq 0$ in V such that*

$$xv = 0 \quad \textit{for all } x \in L.$$

A vector $v \neq 0$ with this property is a *common null vector*.

Here is a preview of the proof, which is an induction on $\dim L$. Choose a proper subalgebra K of L of maximal dimension. The inductive hypothesis, applied to the action of K on L/K, yields a nonzero coset $y + K$ killed by K. This forces K to be an ideal of codimension one. The common null space W of K in V is nonzero and stable under L. The nilpotent restriction of y to W then supplies a vector killed by K and by y, hence by all of L.

Proof. We use induction on $\dim L$, for all nonzero spaces V at once. If $\dim L = 0$, any nonzero v works. Let $d = \dim L \geq 1$. Assume the theorem for subalgebras of dimension less than d in every $\mathfrak{gl}(U)$ with $U \neq 0$.

Step 1: a vector normalising a large subalgebra. The zero subalgebra is proper. Choose a proper subalgebra $K \subsetneq L$ of maximal dimension. Restrict the adjoint representation of L to K. Then L is a K-module and K is a submodule. For $k \in K$, the quotient module $L/K \neq 0$ has the operator

$$\overline{\operatorname{ad}}\, k \colon L/K \to L/K, \qquad y + K \longmapsto [k, y] + K.$$

By Lemma 2.7, $\operatorname{ad} k$ is nilpotent on $\mathfrak{gl}(V)$. Hence it is nilpotent on the invariant subspace L and on the quotient L/K. The image of K in $\mathfrak{gl}(L/K)$ is a subalgebra of dimension at most $\dim K < d$, consisting of nilpotent operators. Apply the inductive hypothesis with $U = L/K$. We obtain a common null vector in L/K. This is a nonzero coset $y + K$, with $y \in L \setminus K$. Being killed by every $\overline{\operatorname{ad} k}$ means that $[k, y] + K$ is the zero coset, so

$$[k, y] \in K \quad \text{for all } k \in K.$$

Step 2: K is an ideal of codimension one. For $k, k' \in K$ and $a, b \in \mathbb{C}$,

$$[k + ay,\ k' + by] = [k, k'] + b[k, y] - a[k', y] \in K.$$

Thus $K + \mathbb{C}y$ is a subalgebra. Its dimension exceeds $\dim K$, so maximality gives $L = K \oplus \mathbb{C}y$. The case $b = 0$ of the display shows $[L, K] \subseteq K$.

Step 3: the common null space of K is L-stable. Let

$$W = \{v \in V : kv = 0 \text{ for all } k \in K\}.$$

The subalgebra K consists of nilpotent maps and has dimension $d - 1$. Induction with $U = V$ gives $W \neq 0$. For $x \in L$, $w \in W$ and $k \in K$, we have $kw = 0$ and $[k, x] \in K$. Hence

$$k(xw) = x(kw) + [k, x]w = 0.$$

Thus $xW \subseteq W$.

Step 4: a null vector of y in W. The restriction of y to W is nilpotent, so it kills some $v \neq 0$ in W. Every $x \in L$ has the form $x = k + ay$ with $k \in K$, so

$$xv = kv + a\,yv = 0.$$

■

A *full flag* in an n-dimensional space V is a chain

$$0 = V_0 \subset V_1 \subset \cdots \subset V_n = V, \qquad \dim V_k = k.$$

A basis is *adapted* to this flag if its first k vectors span V_k. An operator x *lowers* the flag if

$$xV_k \subseteq V_{k-1} \quad (1 \leq k \leq n).$$

Corollary 2.9 (A common flag). *Under the hypotheses of Theorem 2.8, let $n = \dim V$. There is a basis $(v_1, \ldots, v_n)$ of V such that*

$$xv_j \in \operatorname{span}(v_1, \ldots, v_{j-1}) \quad \textit{for all } x \in L,\ 1 \leq j \leq n.$$

Equivalently, the matrix $D(x)$ of each $x \in L$ in this basis is strictly upper triangular.

Proof. Use induction on n. Theorem 2.8 gives $v_1 \neq 0$ with $Lv_1 = 0$. If $n = 1$, we are done. Otherwise $V_1 = \mathbb{C}v_1$ is a submodule of the L-module V. The quotient module $\overline{V} = V/V_1$ gives a homomorphism $L \to \mathfrak{gl}(\overline{V})$, $x \mapsto \bar{x}$. Its image is a subalgebra, and $\bar{x}^m = \overline{x^m} = 0$. By induction, $\overline{V}$ has a basis $(\bar{v}_2, \ldots, \bar{v}_n)$ with

$$\bar{x}\bar{v}_j \in \operatorname{span}(\bar{v}_2, \ldots, \bar{v}_{j-1}).$$

Choose preimages v_j of the $\bar{v}_j$. Then $(v_1, \ldots, v_n)$ is a basis, and $xv_j \in \operatorname{span}(v_2, \ldots, v_{j-1}) + \mathbb{C}v_1$. ■

Corollary 2.10 (Nilpotency of the algebra)**.** *Under the same hypotheses, L is a nilpotent Lie algebra, and $C^{n-1}L = 0$.*

Proof. In the basis of Corollary 2.9, the map $x \mapsto D(x)$ is an injective homomorphism $L \to \mathfrak{gl}_n(\mathbb{C})$ with image in $\mathfrak{n}_n$. Lemmas 2.2(ii) and 2.6 give $C^{n-1}D(L) \subseteq C^{n-1}\mathfrak{n}_n = 0$. Since $L \cong D(L)$, the claim follows. ∎

2.3 The adjoint form

An element $x \in \mathfrak{g}$ is *ad-nilpotent* if $\operatorname{ad} x$ is a nilpotent operator on $\mathfrak{g}$.

Lemma 2.11. *If $C^m\mathfrak{g} = 0$ for some $m \geq 1$, then $(\operatorname{ad} x_1) \cdots (\operatorname{ad} x_m) = 0$ for all $x_1, \ldots, x_m \in \mathfrak{g}$. In particular, every element of a nilpotent Lie algebra is ad-nilpotent.*

Proof. We show $(\operatorname{ad} x_1) \cdots (\operatorname{ad} x_k)(y) \in C^k\mathfrak{g}$ by induction on k. The case $k = 0$ is $y \in C^0\mathfrak{g}$. If $u = (\operatorname{ad} x_2) \cdots (\operatorname{ad} x_k)(y) \in C^{k-1}\mathfrak{g}$, then

$$(\operatorname{ad} x_1)(u) = [x_1, u] \in [\mathfrak{g}, C^{k-1}\mathfrak{g}] = C^k\mathfrak{g}.$$

For $k = m$, the product vanishes. ∎

Theorem 2.12 (Engel's theorem, abstract form)**.** *A Lie algebra $\mathfrak{g}$ is nilpotent if and only if every element of $\mathfrak{g}$ is ad-nilpotent.*

Proof. Lemma 2.11 gives one direction. For the converse, use induction on $\dim \mathfrak{g}$. The zero algebra is nilpotent. Let $\mathfrak{g} \neq 0$, and assume the result in smaller dimension.

The image $\operatorname{ad}\mathfrak{g} \subseteq \mathfrak{gl}(\mathfrak{g})$ of the adjoint representation is a subalgebra consisting of nilpotent operators. Apply Theorem 2.8 with $V = \mathfrak{g} \neq 0$ and $L = \operatorname{ad}\mathfrak{g}$. We obtain $z \neq 0$ such that

$$(\operatorname{ad} x)(z) = [x, z] = 0 \quad \text{for all } x \in \mathfrak{g}.$$

Thus $Z(\mathfrak{g}) \neq 0$.

Let $q\colon \mathfrak{g} \to \overline{\mathfrak{g}} = \mathfrak{g}/Z(\mathfrak{g})$ be the quotient map. For $x, y \in \mathfrak{g}$,

$$(\operatorname{ad} q(x))(q(y)) = q([x, y]) = q((\operatorname{ad} x)(y)).$$

Iterating gives $(\operatorname{ad} q(x))^m \circ q = q \circ (\operatorname{ad} x)^m$. Since q is onto, every element of $\overline{\mathfrak{g}}$ is ad-nilpotent. As $\dim \overline{\mathfrak{g}} < \dim \mathfrak{g}$, the inductive hypothesis makes $\overline{\mathfrak{g}}$ nilpotent. Lemma 2.3(ii) shows that $\mathfrak{g}$ is nilpotent. ∎

Since $\ker \operatorname{ad} = Z(\mathfrak{g})$, we have $\operatorname{ad}\mathfrak{g} \cong \mathfrak{g}/Z(\mathfrak{g})$. The linear theorem was applied to $\operatorname{ad}\mathfrak{g}$, rather than to $\mathfrak{g}$ itself. Since ad has kernel $Z(\mathfrak{g})$, the proof needs the lemma on central quotients to return from $\mathfrak{g}/Z(\mathfrak{g})$ to $\mathfrak{g}$.

2.4 Nilpotent algebras and nilpotent matrices

Let $\rho\colon \mathfrak{g} \to \mathfrak{gl}(V)$ be a representation with $V \neq 0$. Two conditions are easily confused.

(A) The Lie algebra $\mathfrak{g}$ is nilpotent.

(B) Every operator $\rho(x)$ is nilpotent.

Condition (B) says that $\rho(\mathfrak{g})$ satisfies the hypothesis of Theorem 2.8. Then V has a common null vector and a full flag lowered by every $\rho(x)$. Moreover $\rho(\mathfrak{g}) \cong \mathfrak{g}/\ker\rho$ is nilpotent by Corollary 2.10. If ρ is faithful, (B) implies (A). In general, neither condition implies the other.

Example 2.13 (Nilpotent matrices, non-nilpotent algebra)**.** Let ρ be the trivial representation of $\mathfrak{sl}_2$ on $\mathbb{C}^2$. Every $\rho(x) = 0$ is nilpotent, and every nonzero vector is a common null vector. The image $\rho(\mathfrak{sl}_2) = 0$ is nilpotent. The algebra $\mathfrak{sl}_2$ is not, since $C^k\mathfrak{sl}_2 = \mathfrak{sl}_2$.

Example 2.14 (Nilpotent algebra, non-nilpotent matrix)**.** Let $\mathfrak{g} = \mathbb{C}t$ be the one-dimensional abelian algebra. It is nilpotent, and $\operatorname{ad} t = 0$. Define $\rho\colon \mathfrak{g} \to \mathfrak{gl}(\mathbb{C}^2)$ by the matrix $D(t)$ of $\rho(t)$ in the standard basis (e_1, e_2):

$$D(t) = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad D(t)^k = \begin{pmatrix} 1 & 0 \\ 0 & 2^k \end{pmatrix} \neq 0.$$

By linearity, this is a representation, since $[D(t), D(t)] = 0 = D([t,t])$. The operator $\rho(t)$ is not nilpotent. There is no common null vector:

$$D(t)\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} v_1 \\ 2v_2 \end{pmatrix} = 0 \quad \Longrightarrow \quad v_1 = v_2 = 0.$$

The flag $0 \subset \mathbb{C}e_1 \subset \mathbb{C}^2$ is invariant, but $\rho(t)$ does not lower it, since $\rho(t)e_1 = e_1$. No full flag is lowered by $\rho(t)$. Lowering a full flag would make its matrix in an adapted basis strictly upper triangular, hence nilpotent.

The Lie algebra $D(\mathfrak{g}) = \mathbb{C}D(t)$ is abelian, hence nilpotent. So a nilpotent algebra of matrices may contain a matrix that is not nilpotent.

2.5 The strictly upper triangular algebra in dimension four

The algebra $\mathfrak{n}_4$ has basis $E_{12}, E_{13}, E_{14}, E_{23}, E_{24}, E_{34}$. Its general element is

$$x = \begin{pmatrix} 0 & c_{12} & c_{13} & c_{14} \\ 0 & 0 & c_{23} & c_{24} \\ 0 & 0 & 0 & c_{34} \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The product rule for matrix units gives

$$[E_{ij}, E_{kl}] = \delta_{jk}E_{il} - \delta_{li}E_{kj}.$$

The central series. Figure 2.1 places the basis vectors at their matrix positions and marks three brackets. Each bracket adds heights, as in Lemma 2.6. That lemma gives $C^1\mathfrak{n}_4 \subseteq N_2$, $C^2\mathfrak{n}_4 \subseteq N_3$ and $C^3\mathfrak{n}_4 \subseteq N_4 = 0$. The following brackets give the reverse inclusions:

$$[E_{12}, E_{23}] = E_{13}, \qquad [E_{23}, E_{34}] = E_{24}, \qquad [E_{12}, E_{24}] = E_{14}.$$

In the third bracket $E_{24} \in C^1\mathfrak{n}_4$, so $E_{14} \in C^2\mathfrak{n}_4$. Hence

$$\begin{aligned} C^0\mathfrak{n}_4 &= \mathfrak{n}_4, && \text{dimension 6,} \\ C^1\mathfrak{n}_4 &= \operatorname{span}(E_{13}, E_{24}, E_{14}), && \text{dimension 3,} \\ C^2\mathfrak{n}_4 &= \mathbb{C}E_{14}, && \text{dimension 1,} \\ C^3\mathfrak{n}_4 &= 0. \end{aligned}$$

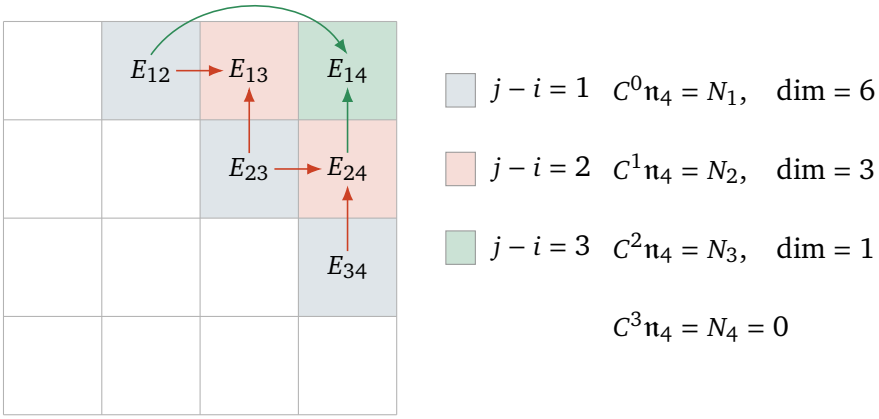


Figure 2.1: The basis of $\mathfrak{n}_4$ at its matrix positions, shaded by the height $j - i$. Two arrows into one entry show a bracket, here $[E_{12}, E_{23}] = E_{13}$, $[E_{23}, E_{34}] = E_{24}$ and $[E_{12}, E_{24}] = E_{14}$, and the heights of the two sources add.

The centre. Since $C^3\mathfrak{n}_4 = 0$, we have $C^2\mathfrak{n}_4 \subseteq Z(\mathfrak{n}_4)$. Conversely, the bracket rule gives

$$\begin{aligned}[E_{12}, x] &= c_{23}E_{13} + c_{24}E_{14},\\ [E_{23}, x] &= c_{34}E_{24} - c_{12}E_{13},\\ [E_{34}, x] &= -c_{13}E_{14} - c_{23}E_{24}.\end{aligned}$$

If x is central, every coefficient except c_{14} vanishes. Thus $Z(\mathfrak{n}_4) = \mathbb{C}E_{14} = C^2\mathfrak{n}_4$.

One step of the Engel induction. For $v = (v_1, v_2, v_3, v_4)^{\mathsf{T}}$,

$$E_{ij}v = v_j e_i.$$

For Step 1, take K spanned by all basis vectors except E_{12}. Since K contains $C^1\mathfrak{n}_4$, we have $[\mathfrak{n}_4, K] \subseteq K$. Its codimension is one, so K is a maximal proper subalgebra. Take $y = E_{12}$. The common null space W of K has $v_3 = 0$, since $E_{13}, E_{23} \in K$. The elements E_{14}, E_{24}, E_{34} force $v_4 = 0$. Hence

$$W = \operatorname{span}(e_1, e_2), \qquad D(y|_W) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \text{in the basis } (e_1, e_2).$$

Step 4 produces $v = e_1$. Indeed, the common null space of $\mathfrak{n}_4$ is exactly $\mathbb{C}e_1$: the units E_{12}, E_{23}, E_{34} force $v_2 = v_3 = v_4 = 0$.

The flag. Put $V_k = \operatorname{span}(e_1, \dots, e_k)$ for $0 \le k \le 4$. For $i < j$,

$$E_{ij}e_l = \delta_{jl}e_i.$$

Thus every $x \in \mathfrak{n}_4$ sends V_k into V_{k-1}.

The adjoint action. Order the basis by decreasing height:

$$\mathcal{B} = (E_{14}, E_{13}, E_{24}, E_{12}, E_{23}, E_{34}).$$

Let $D_{\mathrm{ad}}(x)$ be the matrix of $\operatorname{ad} x$ in this basis. The bracket rule gives

$$D_{\mathrm{ad}}(x) = \begin{pmatrix} 0 & -c_{34} & c_{12} & -c_{24} & 0 & c_{13} \\ 0 & 0 & 0 & -c_{23} & c_{12} & 0 \\ 0 & 0 & 0 & 0 & -c_{34} & c_{23} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

For example, the fourth column comes from

$$[x, E_{12}] = c_{23}[E_{23}, E_{12}] + c_{24}[E_{24}, E_{12}] = -c_{23}E_{13} - c_{24}E_{14}.$$

Every other summand brackets to zero. The matrix $D_{\mathrm{ad}}(x)$ is strictly upper triangular. Its square has only three entries that can be nonzero:

$$(D_{\mathrm{ad}}(x)^2)_{14} = c_{23}c_{34},$$
$$(D_{\mathrm{ad}}(x)^2)_{15} = -2c_{12}c_{34},$$
$$(D_{\mathrm{ad}}(x)^2)_{16} = c_{12}c_{23}.$$

Its cube is zero, in agreement with $C^3\mathfrak{n}_4 = 0$. Thus the algebra has explicit lowered flags in its defining and adjoint modules. These modules have dimensions four and six. The common null space in the adjoint module is its centre $\mathbb{C}E_{14}$.

Self-checks

(1) Is the upper triangular algebra $\mathfrak{b} \subseteq \mathfrak{gl}_2(\mathbb{C})$ nilpotent? Does Theorem 2.8 apply to it?

(2) The trivial representation of $\mathfrak{sl}_2$ on a nonzero space sends every element to a nilpotent operator. What does Theorem 2.8 conclude, and what does it not conclude?

(3) Find $x \in \mathfrak{n}_4$ with $(\operatorname{ad} x)^2 \neq 0$. Why is $(\operatorname{ad} x)^3 = 0$ for every $x \in \mathfrak{n}_4$?

Answers. (1) No, since $C^k\mathfrak{b} = \mathbb{C}e$ for all $k \geq 1$. The theorem does not apply, because $I_2 \in \mathfrak{b}$ is not nilpotent. (2) The theorem gives a common null vector. The nilpotency corollary shows that the image $\rho(\mathfrak{sl}_2) = 0$ is nilpotent. The representation is not faithful. Neither result implies that $\mathfrak{sl}_2$ is nilpotent. (3) Take $x = E_{12} + E_{23}$. Then

$$(\operatorname{ad} x)E_{34} = E_{24}, \qquad (\operatorname{ad} x)E_{24} = E_{14} \neq 0.$$

Since $C^3\mathfrak{n}_4 = 0$, Lemma 2.11 gives $(\operatorname{ad} x)^3 = 0$.

Summary

The lower central series records iterated brackets. Subalgebras and quotients of nilpotent algebras are nilpotent. A nilpotent central quotient forces nilpotency.

Engel's linear theorem gives a Lie algebra of nilpotent operators on a nonzero space a common null vector and a lowered full flag. Applied to $\operatorname{ad}\mathfrak{g} \cong \mathfrak{g}/Z(\mathfrak{g})$, it shows that a Lie algebra is nilpotent exactly when all its elements are ad-nilpotent. A nilpotent algebra may still act by non-nilpotent matrices. Lecture 3 replaces null vectors by common eigenvectors. Lecture 4 uses the linear form in the sketch of a trace criterion for solvability.

Reading

Humphreys, *Introduction to Lie Algebras and Representation Theory*, Sections 3.2–3.3; Section 3.1 belongs with Lecture 3. Erdmann–Wildon, *Introduction to Lie Algebras*, Chapters 4–6. Exercise Sheet 2 treats centres, quotients and the limits of the matrix criterion.